9.3.8

AI25BTECH11017-BALU

Question:
Find the area of the region $\{(x, y) : x^2 \le y \le x\}$.
Solution:
Let us solve the given equation theoretically and then verify the solution computationally
According to the question,
equation line

$$\mathbf{x} = \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{0.1}$$

equation of parabola is

$$\mathbf{x}^T \mathbf{v} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \tag{0.2}$$

where

$$\mathbf{v} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{0.3}$$

$$\mathbf{u} = \begin{pmatrix} 0 \\ \frac{-1}{2} \end{pmatrix} \tag{0.4}$$

$$f = 0 \tag{0.5}$$

substute 0.1 in 0.2
$\lambda = 0, 1$ so intersection points are

$$\mathbf{a_1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{0.6}$$

$$\mathbf{a_2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{0.7}$$

hence the requried area is

$$\int_0^1 (x - x^2)\, dx = \frac{1}{6} \tag{0.8}$$

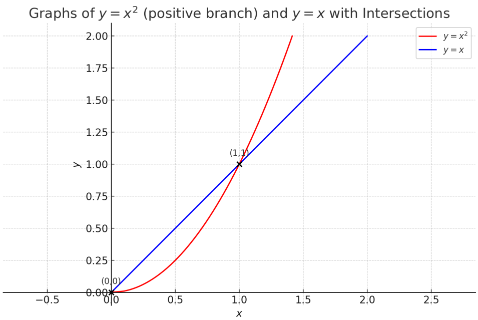
Graphs of $y = x^2$ (positive branch) and $y = x$ with Intersections
$y = x^2$
$y = x$
(1,1)
(0,0)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
x
y

Fig. 0.1